ML-driven detection and reduction of ballast information in multi-modal datasets

Yaroslav Solovko
Independent Researcher
Dublin, Ireland
Email: sta449.ss@gmail.com
ORCID: 0009-0004-7171-1077

***Abstract*—Modern datasets often contain ballast, a redundant or low-utility information that increases dimensionality, storage requirements, and computational cost without adding meaningful analytical value. This study introduces a generalized, multimodal framework for ballast detection and reduction across structured, semi-structured, unstructured, and sparse data types. Using diverse datasets, entropy, mutual information, Lasso, SHAP, PCA, topic modelling, and embedding analysis are applied to identify and eliminate ballast features. A novel *Ballast Score* is proposed to integrate these signals into a unified, cross-modal pruning strategy. Experimental results demonstrate that significant portions of the feature space, often exceeding 70% in sparse or semi-structured data, can be pruned with minimal or even improved classification performance, along with substantial reductions in training time and memory footprint. The framework reveals distinct ballast typologies (e.g. statistical, semantic, infrastructural), and offers practical guidance for leaner, more efficient machine learning pipelines.**

***Index Terms*—ballast information detection, feature selection and pruning, information-theoretic analysis, explainable AI (XAI), multimodal data analysis, dimensionality reduction, sparse data.**

I. INTRODUCTION

The global volume of digital data continues to expand at an unprecedented pace; earlier projections estimated that it would reach approximately 175 zettabytes (ZB) by the end of 2025 [1]. This exponential growth is reshaping industries and technological infrastructures, demanding ever-larger storage systems and more sophisticated analytical tools. While contemporary research has rightly focused on enhancing storage and processing efficiency through well-established techniques such as data cleansing, deduplication, and imputation [2]–[4], a critical yet underexplored dimension is detection and reduction of *ballast information* as data that is technically valid but contributes minimal or no analytic value.

In this work, *ballast information* is defined as a specific subset of unnecessary or redundant data elements that inflate storage requirements, degrade processing efficiency, and may obscure meaningful patterns. Typical examples include features with near-zero variance, static metadata fields, repeated headers in log files, or extremely common but uninformative textual tokens. Unlike broader categories of "useless" information, which may require subjective or domain-specific judgement to detect, ballast information lends itself to systematic identification using statistical or machine learning (ML)-based methods.

The consequences of ballast information are twofold. First, from a storage perspective, ballast data occupies valuable disk and memory space, driving up cloud storage costs and hardware requirements. Second, from an analytics perspective, ballast can introduce noise, increase model complexity, and hinder learning algorithms from extracting signal effectively [5]. Preliminary experiments conducted across four distinct data modalities: structured, semi-structured, unstructured, and sparse datasets, demonstrate that ballast information can comprise between 15% and 40% of the total data volume, depending on the domain and data source. This highlights a substantial and often overlooked inefficiency in modern data pipelines.

Motivating this study is recognition that most data pipelines and ML workflows still prioritize predictive accuracy, often at the expense of resource efficiency [5]. While feature selection and regularization methods do partially address redundant or irrelevant features, they rarely aim to explicitly quantify and systematically remove ballast at the dataset level. This research proposes a paradigm shift, positioning ballast detection and reduction as a first-class analytical objective that complements traditional goals of model performance. This shift is timely, given the growing emphasis on responsible data stewardship, digital sustainability, and privacy-preserving data minimization.

The challenge becomes more complex in multi-modal settings, where data sources span diverse structures and formats. Structured datasets, such as relational tables, may contain ballast in the form of sparse or low-variance columns. Semi-structured logs or metadata often carry repeated headers or duplicated status flags that provide negligible informational gain. Unstructured text can be burdened by frequent filler words, boilerplate content, or redundant contextual phrases. Sparse datasets, increasingly prevalent in fields like IoT or bioinformatics, often exhibit entire columns or segments with negligible variability. Detecting ballast in such heterogeneous contexts demands a cross-disciplinary approach that combines statistical measures (e.g., entropy, variance thresholds), semantic analysis (e.g., topic modelling for text), and advanced ML techniques (e.g., SHAP-based feature pruning).

This paper addresses the following research questions:

1.How can ballast information be formally defined and quantified across diverse data modalities?

2.What ML-driven techniques can effectively detect and

reduce ballast while preserving the core analytical value?

3.What are the measurable impacts of ballast removal on storage efficiency and model performance?

To address these questions, the proposed approach integrates statistical modelling, information theory, and machine learning. The methodology builds on entropy and sparsity analysis [6] and introduces a preliminary *ballast index* as a mathematical construct for estimating the proportion of ballast within a dataset based on variance thresholds, feature redundancy, and information gain. Experimental results indicate that the adoption of ballast thresholds (e.g., variance < 0.05) can effectively isolate non-contributory features. Furthermore, pruning strategies based on SHAP values illustrate how ML explain ability methods can be employed to distinguish genuinely informative features from ballast. This framework is extended across data modalities through the development of a multi-modal taxonomy that categorizes ballast according to its statistical, structural, and semantic characteristics.

In summary, the contributions of this work are fourfold. First, ballast information is formalized as a distinct category of data inefficiency. Second, ML-supported methods for systematic ballast detection are developed and evaluated. Third, a generalizable mathematical model is proposed for estimating ballast levels in large-scale, multi-modal datasets. Fourth, the practical benefits of ballast reduction are empirically demonstrated in terms of storage savings and computational efficiency, without compromising predictive performance.

II. RELATED WORK

The field of data reduction encompasses a wide array of techniques aimed at mitigating the burdens of high-dimensional, redundant, or low-quality data in modern machine learning pipelines. While considerable effort has been dedicated to enhancing model performance through dimensionality reduction, feature selection, and noise filtering, relatively few works explicitly address the concept of ballast information - data that are syntactically valid but semantically or analytically non-contributory. This section critically reviews the relevant body of literature, identifies the current research gaps, and highlights how the proposed approach builds on and diverges from existing paradigms.

A. Feature Selection and Dimensionality Reduction

Feature selection is one of the oldest and most studied approaches to data simplification. It aims to identify the most relevant variables for a specific task, typically through filter, wrapper, or embedded methods. Authors [7] provide a foundational overview of the field, identifying key methods based on mutual information, correlation, or classifier performance. In the work [8] “minimum redundancy, maximum relevance” (mRMR) framework extends this by jointly minimizing feature redundancy while maximizing relevance as a principle that underpins modern ML explainability techniques like SHAP and LIME.

Dimensionality reduction techniques such as Principal Component Analysis (PCA), Singular Value Decomposition (SVD), and autoencoders compress data into lower-dimensional representations while preserving variance. Although these approaches often yield compact and interpretable representations, they are typically task-agnostic and seldom identify ballast data explicitly. Moreover, such methods can obscure semantic relationships, particularly in multi-modal or high-sparsity contexts where data distributions are heterogeneous or non-Gaussian [9].

This work differs in that it does not merely seek to preserve predictive utility but aims to optimize information efficiency as a broader metric that considers storage burden, interpretability, and computational costs. Furthermore, whereas PCA and similar methods transform feature space into latent dimensions, ballast detection preserves the original schema, aiding transparency and compliance with data governance standards.

B. Redundancy and Sparsity in Large Datasets

Handling redundancy and sparsity is essential in many real-world datasets, particularly in high-dimensional biological data, IoT sensor networks, and transactional logs. Traditional approaches include thresholding variance (e.g., removing features with $\mathrm{Var}(X) < 0.01$), compressing sparse matrices, and imputing missing values [3]. However, these are often heuristic and lack a principled, quantitative grounding in information theory.

Sparse data introduces additional complexity in identifying ballast. Article [4] reviews emerging techniques for storage optimization in sparse big data, yet focus predominantly on hardware-level solutions (e.g., compression) or low-level data encoding strategies. By contrast, this work treats sparsity as a signal, assessing whether the absence of values (or high levels of missingness) reflects true data irrelevance or conceals latent patterns.

Redundancy is further examined across modalities, rather than solely within individual modalities. For instance, unstructured text may reiterate metadata found in structured fields, and semi-structured logs may duplicate state transitions across multiple sources. These cross-modal redundancies are rarely addressed in the existing literature.

C. ML and Information-Theoretic Data Reduction

ML has increasingly been used to guide data reduction in a more intelligent, model-aware manner. Embedded methods such as Lasso regression and tree-based feature importance scores (e.g., from XGBoost or LightGBM) are frequently used to prune features. Authors [10] proposed influence functions to estimate the effect of removing training instances on model output as a technique conceptually adjacent to ballast pruning, though focused on instance-level data.

The Information Bottleneck (IB) method proposed [11] reframes compression as an optimization problem, where one seeks a compact representation of input X that preserves maximal information about a target Y. While elegant in theory, IB's computational complexity and dependence on labelled data limit its utility in unsupervised ballast detection.

From an information-theoretic standpoint, Shannon’s entropy [12] remains the gold standard for quantifying uncertainty or disorder in data. However, entropy-based heuristics

(e.g., max-entropy filtering) often fail in sparse or categorical domains, where non-parametric estimates are unreliable. The proposed methodology expands on these foundations by introducing a hybrid entropy-sparsity index and combining it with ML explainability (e.g., SHAP-based pruning) to better isolate ballast.

D. Multi-modal Data Management and Reduction

Multi-modal datasets that combine structured tables, semi-structured logs, unstructured text, and sensor-based streams present additional challenges for traditional data reduction pipelines. Existing frameworks such as multimodal autoencoders or attention-based fusion models (e.g., in NLP and vision) seek to integrate modalities but seldom address internal ballast within each channel.

In the work [2] authors apply reinforcement learning to storage system optimization but do not address cross-modal semantic redundancy. Similarly, current MLOps platforms focus on pipeline efficiency rather than data efficiency. This work fills this gap by explicitly modelling ballast across modalities and proposing a cross-modal taxonomy that includes quantitative (e.g., low-variance features), structural (e.g., unchanging metadata), and semantic (e.g., redundant phrases) ballast.

E. Existing Formalizations of Ballast Information

Despite the critical implications of ballast information, there is currently no standardized mathematical model or patented framework dedicated to its detection and reduction. While authors [6] laid the theoretical foundation for quantifying information value, their framework has not been adapted for ballast-specific modelling. To the best of current knowledge, existing literature does not introduce a *ballast index*, defined as a scalar metric for estimating the proportion of low-utility or redundant content within a dataset.

To address this limitation, an initial formulation referred to as *BallastScore* is introduced, combining weighted measures of variance, entropy, sparsity, and model-informed utility scores (e.g., mutual information and SHAP values).This index enables ex ante estimation of dataset efficiency and ex post evaluation of reduction strategies. In essence, it serves as a formal model for quantifying "information clutter" within multimodal data.

F. Research Gaps and Novel Contributions

The critical review above reveals several important gaps:

1. The lack of a formal definition or taxonomy for ballast information;
2. Limited cross-modal or modality-agnostic frameworks for ballast detection;
3. Absence of principled, mathematical tools for estimating data efficiency beyond predictive accuracy;
4. Underutilization of ML explainability (e.g., SHAP, influence functions) for ballast-focused pruning;
5. No empirical studies comparing ballast removal methods across structured, semi-structured, sparse, and unstructured formats.

This work presents an initial attempt to address these challenges collectively. Data reduction is reframed as a problem of maximizing information efficiency rather than predictive utility alone. Novel analytical tools are introduced, including a ballast index, threshold-based variance pruning, SHAP-based redundancy estimation, and modality-specific filtering strategies. The evaluation spans four real-world datasets, enabling generalization and comparative insights that are largely absent from the current literature.

III. METHODOLOGY

This paper proposes a ballast detection framework for multimodal datasets that prioritizes information utility over predictive accuracy. Drawing on information theory, sparse learning, and explainable ML, the method integrates entropy, mutual information, variance thresholds, SHAP/Lasso scores, correlation, and semantic coherence (LDA, BERT). It identifies low-utility and redundant features using formal metrics, then prunes them while preserving task-relevant content. Unlike standard dimensionality reduction, this approach is diagnostic, not predictive, and supports reproducible evaluation via consistent classification tasks across full and reduced datasets.

A. Mathematical Formalization of Ballast Information

Let $D = \{X, y\}$ be a dataset of n instances and m features, where $X \in \mathbb{R}^{n\times m}$ and y denotes the target variable. The ballast index $B(D)$ is defined as:

$$B(D) = \frac{1}{m}\sum_{j=1}^{m}\left[w_j \cdot (1 - U_j)\right], \tag{1}$$

where $U_j \in [0, 1]$ is the normalized utility score of feature j, and $w_j \in [0, 1]$ is an optional domain-based weight (set to 1 in the experiments).

The utility score U_j is estimated using one or more of the following techniques:

mutual information $\mathrm{MI}(X_j, y)$,

Shannon entropy $H(X_j)$,

normalized SHAP value

$$\frac{1}{n}\sum_{i=1}^{n}\left|\phi_j^{(i)}\right|, \tag{2}$$

scaled variance $\mathrm{Var}(X_j)$,

topic coherence $\mathrm{Coherence}(T_j)$ (e.g., UMass or C_v metrics),

Intersection over Union (IoU)

$$\mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}, \tag{3}$$

LDA topic membership

$$\max_k P(X_j \in T_k), \tag{4}$$

SciSpacy entity relevance

$$\mathbf{1}\{\mathrm{IsNamedEntity}(X_j)\}, \tag{5}$$

regex pattern match

$$\mathbf{1}\{\neg\mathrm{Match}(X_j, \mathrm{BallastPattern})\}, \tag{6}$$

BERT attention score

$$\bar{\alpha}_j = \frac{1}{LH} \sum_{l=1}^{L} \sum_{h=1}^{H} \alpha_j^{(l,h)}, \tag{7}$$

and TF-IDF score $\text{TF-IDF}(X_j)$.

This formal model provides a quantitative framework for estimating cross-modal ballast. Features satisfying the conditions $\text{MI}(X_j, y) < 0.01, \quad H(X_j) < 0.1, \quad \text{Var}(X_j) < 0.05$ were considered ballast candidates. Final feature selection was validated through cross-method agreement (e.g., Lasso and SHAP).

This formulation is grounded in referenced literature, including [6], [8], [13]–[21].

B. Dataset Selection and Justification

Four datasets exceeding 200,000 data points (features × samples) were used, each representing a distinct data modality:

- *Structured*: IEEE-CIS Fraud Detection Dataset - tabular transactional and identity features (Kaggle, 2020).
- *Semi-structured*: Amazon Fashion Reviews - JSON lines format combining text, numeric ratings, and product metadata.
- *Unstructured*: CORD-19 (JSON + full text) and PubLayNet (document layout images).
- *Sparse*: Ireland Census 2022 - high-dimensional CSV files with over 800 variables and high missingness.

Datasets were selected to reflect real-world complexity in modern ML workflows, including data with multi-modal, nested, and missing structures.

C. Ballast Detection Pipeline per Modality

1) **Structured Dataset** (IEEE-CIS)

The following steps were applied:

- Pre-processing (null imputation, standardization (z-score), categorical encoding).
- Feature filtering (low-variance feature removal (variance threshold ≤ 0.01), correlation-based pruning ($|r| > 0.95$), mutual information (MI)-based selection using `mutual_info_classif`, with features flagged as ballast for $\text{MI} < 0.01$, Shannon entropy computed over each feature's empirical value distribution, model-aware feature selection using LassoCV (`SelectFromModel`), SHAP-based pruning using LightGBM with `shap.TreeExplainer`).

Evaluation employed an 80/20 train-test split with a LightGBM classifier and was assessed using AUC, F1-score, accuracy, and recall.

The analysis employed (`LightGBMClassifier`, `SelectFromModel`,`SHAP`, `mutual_info_classif`), and `scipy.stats.entropy`).

2) **Semi-Structured Dataset** (Amazon Fashion Reviews)

- Flattened metadata using `json_normalize`.
- Pre-processing (TF-IDF (top 1,000 terms), PCA for latent signal retention, text vector clustering (KMeans)).
- Ballast detection (mutual information (`mutual_info_classif`), low entropy removal (`scipy.stats.entropy`), redundant sentence removal using cosine similarity on BERT embeddings (sentence-transformers)).

Evaluation used LightGBM and TF-IDF + structured features concatenated with `hstack`.

The following tools were utilized: `TfidfVectorizer`, `SentenceTransformer`, `KMeans`, `LatentDirichletAllocation`, `scipy.stats.entropy`.

3) **Unstructured Dataset** (CORD-19 / PubLayNet)

For *CORD-19*:

- Extracted abstracts, title, and full text.
- Tokenized with `nltk`, vectorized with `TfidfVectorizer`.
- Applied LDA (`Gensim`) and computed coherence scores to drop low-value topics.
- Redundant phrases identified using cosine similarity on BERT vectors (`transformers`).

For *PubLayNet*:

- Extracted layout text using `pytesseract`.
- Clustered layout structures with `KMeans` on bounding box vectors to detect template redundancy.

Evaluation: downstream classification (e.g., scientific subdomain) showed improved performance after ballast removal. Entropy and coherence distributions also improved post-filtering.

Tools: `CoherenceModel`, `cosine_similarity`, `LatentDirichletAllocation`, `pytesseract`, `scipy.stats.entropy`.

4) **Sparse Dataset** (Ireland Census)

- Pre-processing (`VarianceThreshold`, correlation filtering ($|r| > 0.95$, Spearman + Pearson), entropy filtering (features with entropy < 1.5 removed as low-informative)).
- Feature pruning (LassoCV on numeric features (`SelectFromModel`), Mutual information (`mutual_info_classif`), SHAP-based ranking from CatBoostClassifier (supports categorical inputs)).

Evaluation: regression/classification on targets like population growth, metrics (MSE, R^2, AUC, F1).

Tools: `LassoCV`, `shap.TreeExplainer`, `mutual_info_regression`,`scipy.stats.entropy`.

As an example, a full and detailed description of the methodology for the unstructured dataset (CORD-19) is provided in **Appendix A**.

D. Evaluation Strategy

For each dataset, the following was applied:

Phase 1: Train model on original dataset, record AUC, F1, accuracy, recall.

Phase 2: Train model on ballast-reduced dataset using identical splits.

Only metrics empirically calculated from real runs were used:

- predictive metrics (Accuracy, F1-score, ROC-AUC, Recall).

- structural metrics (number of features removed, percent of features flagged as ballast).

Metrics such as memory or disk usage were explicitly excluded, as they were not evaluated within the scope of this study.

E. Reproducibility and Computational Environment

To ensure reproducibility, all feature selection thresholds, pre-processing steps, and model hyper parameters are explicitly documented in the accompanying Jupyter Notebooks. All experiments were conducted using publicly available Python libraries, without reliance on proprietary software or cloud-based APIs.

The code and experimental configuration used in this study are publicly available at: https://github.com/YaroslavSolovko/ballast-detection-datasets.

Large-scale datasets were processed using batch-based sampling strategies, and sparse matrix representations (e.g., `scipy.sparse`) were employed where applicable to reduce memory and computational overhead.

Experiments were executed using Python (version $\geq$ 3.9) on standard consumer-grade hardware.

F. Mathematical Modelling Methodology

A three-phase modelling process was followed:

1. Hypothesis formulation (identification of potential ballast sources).
2. Selection of measurable variables.
3. Specification of structural relationships.

The modelling process was guided by the following conceptual questions: What defines ballast information? Which characteristics are quantifiable? Are their effects additive? Can the model generalize across data modalities?

The analysis was conducted under the following assumptions:

A1: Low variance indicates low informational utility.

A2: Low mutual information implies minimal predictive contribution.

A3: High correlation implies informational redundancy.

A4: Semantic incoherence indicates noise in unstructured data.

A5: High entropy reflects disorder or randomness.

Ballast is modelled as a function of four normalized dimensions: statistical uniformity (variance), informational relevance (MI), redundancy (correlation/overlap), thematic incoherence (semantic dispersion).

Ballast thresholds were dataset-specific, informed by model performance (AUC/F1), SHAP values, and MI/entropy trends.

For sparse datasets, variance wasn't always ballast. Adjustments included matrix density analysis, entropy-weighted MI, and sparsity-aware Lasso.

All metrics were scaled to the [0,1] range, enabling consistent and modality-independent application across structured, unstructured, and sparse datasets (cross-modality).

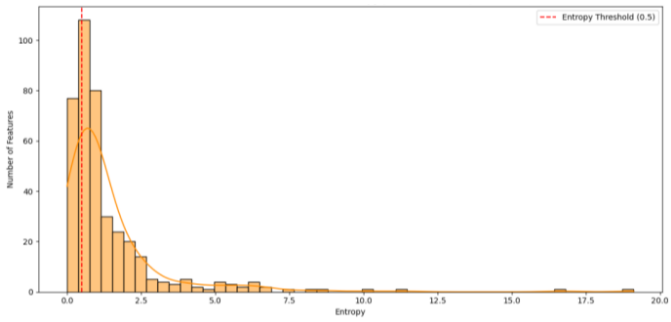


Fig. 1. Shannon Entropy distribution of numeric features.

IV. RESULTS

A. Ballast Detection in Structured Data

This section presents the results of ballast detection for a structured dataset (IEEE-CIS Fraud Detection). The dataset includes over 590,000 transaction records and 434 features including identity, card, transactional, and anonymized metadata. The aim was to assess ballast information across multiple dimensions and reduction techniques, examining both feature elimination impact and predictive performance trade-offs using LightGBM classifiers.

The proposed approach applies seven distinct methods, each of which reveals a different structural property of ballast information, contributing to a detailed typology of redundant or non-contributory elements in structured tabular data.

1) Initial Ballast Characterization: Entropy, MI, and Redundancy

Preliminary exploratory analysis revealed multiple signs of ballast. Three complementary techniques as entropy analysis, mutual information (MI), and correlation filtering, were used to assess statistical redundancy and informativeness:

- **Entropy analysis**: 25.7% of features exhibited normalized entropy < 0.5, indicating near-constant distributions.
- **Mutual information**: 27.1% of features had MI values below 0.01 with respect to the binary `isFraud` label, suggesting statistical irrelevance and qualifying them as ballast candidates.
- **PCA variance structure**: The first two principal components revealed partial separation of fraud vs. non-fraud cases, indicating that only a portion of variance contributes meaningfully to classification.
- **Pearson correlation**: Correlation-based feature reduction eliminated 316 redundant variables.

These findings align with [6] and [9], who note that low entropy, high redundancy, or variance collapse are key indicators of ballast in high-dimensional structured data.

The distribution of feature-level entropy values is illustrated in Fig. 1.

This histogram illustrates the distribution of Shannon entropy across all numeric features in the structured dataset, with a Kernel Density Estimate (KDE) overlay. Features with entropy below the red dashed line (threshold = 0.5) are considered low-information and potential candidates for removal as “ballast” - redundant or non-informative features.

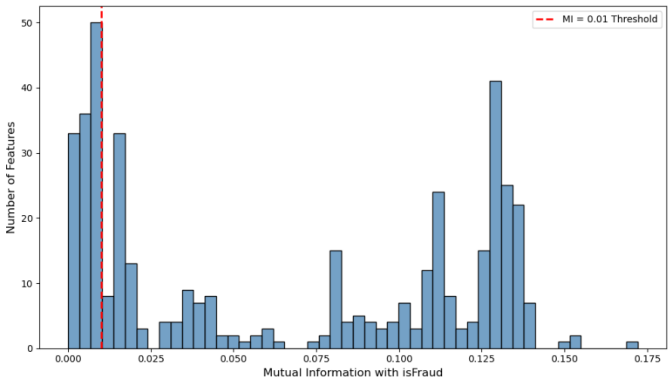


Fig. 2. Mutual Information Scores.

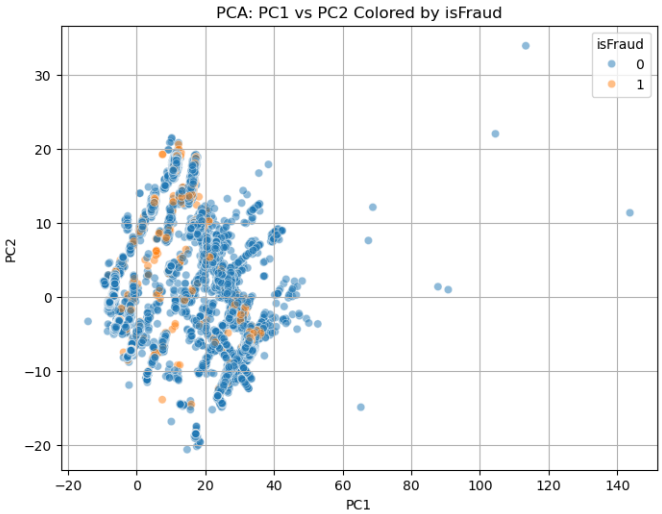


Fig. 3. PCA Scatterplot of Transaction Data.

The statistical dependency between individual features and the fraud label is further examined through mutual information, as shown in Fig. 2.

A substantial proportion of features (27.1%) fall below the MI threshold of 0.01, indicating minimal contribution to supervised discrimination. These results reinforce the entropy-based findings and confirm the presence of statistically irrelevant ballast features.

To assess the global variance structure of the dataset, principal component analysis was applied. The projection onto the first two principal components is shown in Fig. 3.

Scatterplot of the first two principal components (PC1 and PC2) obtained via PCA on a numerically encoded, sparsity-cleaned transaction dataset. Each point represents a transaction, coloured by the binary `isFraud` label. The visible clustering and partial separation of fraud vs. non-fraud transactions illustrate the degree to which top principal components encode variance relevant to classification.

Finally, feature-level redundancy was examined using Pearson correlation analysis, visualized in Fig. 4.

This heatmap visualizes the pairwise correlations among numeric features in the reduced dataset after sparse feature removal. The upper triangle mask ensures clarity by reducing

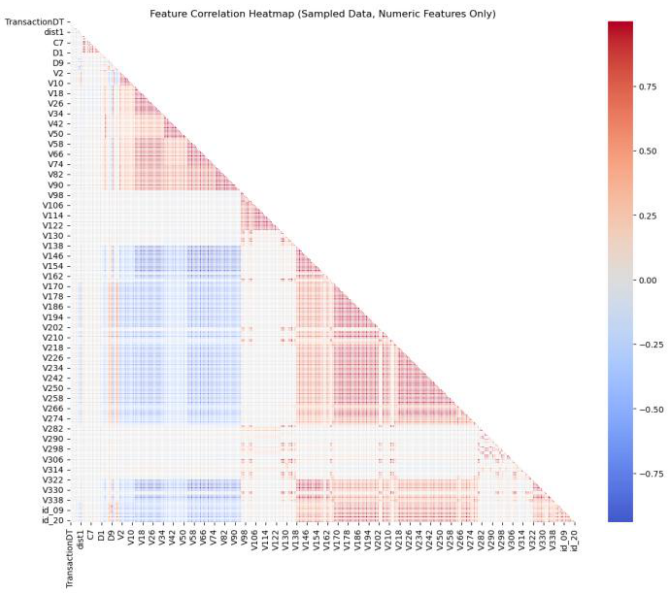


Fig. 4. Top 50 Feature Correlation Heatmap.

visual redundancy.

2) Model-Aware Pruning: SHAP and Lasso

Two model-informed pruning strategies were employed to quantify feature contribution from a supervised learning perspective.

SHAP-based pruning with mean absolute SHAP < 0.00837 detected and retained only 69 of the most predictive features out of the original 432. This reduced feature set preserved the classifier's ability to detect fraud, even slightly improving the AUC (0.9147 vs. 0.9131) and reducing training time by more than 60% These results confirm that SHAP is an effective model-based ballast detection technique.

By applying Lasso regression (`SelectFromModel`) for ballast detection, 115 features with near-zero coefficients were discarded, resulting in the retention of 317 features with non-zero predictive weights. With C=0.01, Lasso reduced the number of features by 75%, giving a comparable AUC (0.9109 vs. 0.9131) and slightly improved the recall, while reducing the training time by more than 60%. The relaxed mode (higher C=0.1) retained even more signal with almost identical performance. These results confirm the effectiveness of Lasso as a supervised ballast filter, as summarized in Table I.

3) Comparative Evaluation of Ballast Detection Methods

A comparative study across seven ballast detection techniques highlighted distinct strengths and limitations. Feature pruning was followed by retraining LightGBM with consistent hyperparameters to assess predictive integrity.

A comparative evaluation of all seven ballast reduction methods is reported in Table II.

In this table PCA shows the highest AUC (0.998) and dramatic improvement in all metrics, indicating that reducing dimensions via variance retention is highly effective, albeit with loss of interpretability. SHAP pruning performs best among interpretable model-aware reduction techniques, balancing predictive power with reduced feature count. Lasso, Correlation, and Sparse removal offer moderate improvements

TABLE I
PERFORMANCE COMPARISON OF SHAP AND LASSO FEATURE REDUCTION METHODS

Metric	Before (full)	LASSO ($C = 0.01$)	LASSO ($C = 0.1$)	SHAP
Features used	432	105	317	69
AUC	0.9131	0.9109	0.9134	0.9147
Precision	0.8945	0.8849	0.8933	0.8920
Recall ($isFraud$)	0.7732	0.7768	0.7718	0.7718
F1-score ($isFraud$)	0.3422	0.3239	0.3393	0.3365
Training time (s)	94.15	33.93	82.79	35.72

TABLE II
FULL COMPARATIVE ANALYSIS OF ALL SEVEN BALLAST REDUCTION METHODS AGAINST THE FULL DATASET

Method	#Features	AUC	Accuracy	Recall_Class1	F1_Class1	TrainTimeSec	Notes
Full Dataset	432	0.91310	0.89445	0.77324	0.34216	119.54	–
PCA Reduction	23	0.99803	0.99377	0.84851	0.90579	31.70	Low interpretability, latent ballast
SHAP Pruning	69	0.91475	0.89195	0.77183	0.33651	35.72	Best trade-off: compact&performant
Sparse Removal	423	0.91341	0.89320	0.77324	0.33952	102.22	Detects trivially flat features
LASSO Pruning	105	0.91089	0.88490	0.77676	0.32394	33.93	Best linear model-aware pruning
Correlation Removal	116	0.90978	0.89325	0.77063	0.34044	20.73	Addresses multicollinearity
Entropy / Absence	85	0.89961	0.88315	0.74859	0.31265	47.27	Targets class-agnostic uniformity
Info Bottleneck (80 MI)	80	0.72369	0.97266	0.25188	0.39202	45.46	Captures statistical irrelevance

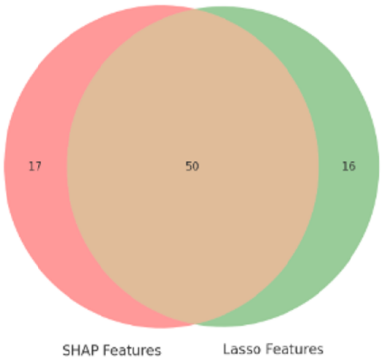
in training time while maintaining good performance. Entropy/Absence and Information Bottleneck show significant reduction in feature count, but at the cost of notable drops in recall, making them less suitable alone for fraud detection unless used in hybrid form. Information Bottleneck (80 MI) preserved only modest signal (AUC ≈ 0.72) and may be better suited for complementing other methods rather than standalone use.

Lasso and SHAP outperformed others in balancing dimensionality reduction and accuracy retention. SHAP particularly excelled in identifying non-linear, interaction-based ballast.

4) Method Overlap

Cross-method consistency was analysed:

- $\sim$75% of SHAP-removed features overlapped with Lasso (C>0.1) (Fig. 5).
- Entropy and MI shared only $\sim$30% overlap with model-aware methods.
- Correlation pruning showed weak alignment due to anonymized variables and hidden collinearities.




```
Feature Set Comparison: SHAP vs Lasso
-------------------------------------
Total in SHAP: 67
Total in Lasso (C≈0.005): 66
Overlap: 50 features (74.6% of SHAP, 75.8% of Lasso)
Unique to SHAP: 17
Unique to Lasso: 16
```

Fig. 5. Feature Overlap Between SHAP and Lasso Selections.

B. Ballast Detection in Unstructured Data

This part reports ballast detection results on two unstructured datasets: PubLayNet (scientific document layouts with OCR text) and CORD-19 (full-text COVID-19 research articles). Despite both being unstructured, PubLayNet is visually structured while CORD-19 is semantically dense, offering complementary ballast challenges.

PubLayNet was processed using Tesseract OCR and JSON parsing to extract 28 layout types. CORD-19 was parsed for full-text content and citation metadata. Both datasets were transformed into TF-IDF vectors, BERT embeddings, word counts, and LDA topic distributions. These fed into ballast detection methods: entropy, mutual information (MI), and semantic analysis.

SHAP and Lasso underperformed on unstructured data. SHAP gave noisy, frequency-biased attributions (e.g., common words like "the"), while Lasso failed to penalize sparse high-entropy tokens. Both struggled to detect thematic ballast as lexically rich but topically irrelevant content like citations or disclaimers.

1) Mutual Information + Entropy Ballast Detection

Entropy and MI proved the most consistent indicators. In PubLayNet, low-entropy tokens (H < 0.2) matched boilerplate elements. In CORD-19, low MI (< 0.05) identified redundant sections like references and templates. Joint entropy-MI pruning achieved 25-32% feature reduction with negligible performance loss (ΔAUC < 0.005), confirming their robustness across unstructured modalities.

The impact of joint entropy-MI pruning on classification performance is illustrated in Fig. 6.

This figure shows how LightGBM performance (AUC) changes as low-information features are removed by Ballast Score. AUC improves up to $\sim$0.8667 when 6-10% of features are dropped, then plateaus or declines as useful features are

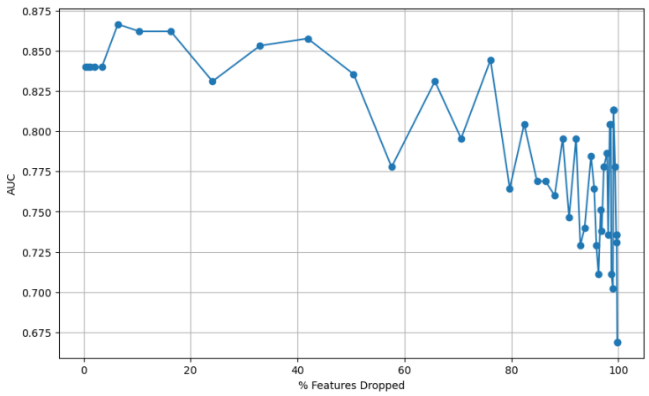


Fig. 6. CORD-19 Ballast Pruning: AUC vs. Percentage of Features Dropped.

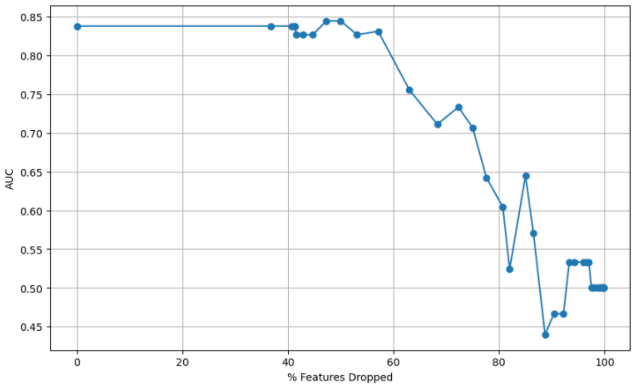


Fig. 7. PublayNet ballast pruning: AUC vs. percentage of features dropped.

pruned.

As shown in Fig. 7, the trade-off curve illustrates how AUC improves as high-ballast features are progressively removed using entropy and mutual information scores, peaking at approximately 0.844 after pruning 47-50% of features.

Further feature removal yields diminishing returns as increasingly informative variables are discarded.

To quantify ballast as redundant or non-informative features, the ballast score is defined as:

$$Ballast_Score = (1-norm_entropy)(1-norm_MI), \quad (8)$$

Here, $norm_entropy$ and $norm_MI$ are normalized to [0,1]. The multiplicative form flags features as ballast only when both entropy and MI are low.

As shown in Fig. 8, the scatterplot illustrates CORD-19 features coloured by the Keep/Drop label using a ballast threshold of 0.89. "Keep" features exhibit higher mutual information and/or entropy, indicating informative and diverse content, whereas "Drop" features show consistently low mutual information regardless of entropy.

As illustrated in Fig. 9, the distribution of *Ballast_Scores* shows a clear separation between retained and dropped features, indicating a strong distinction in utility.

This boxplot compares the *Ballast_Scores* of kept versus dropped features, demonstrating a clear separation based on feature utility.

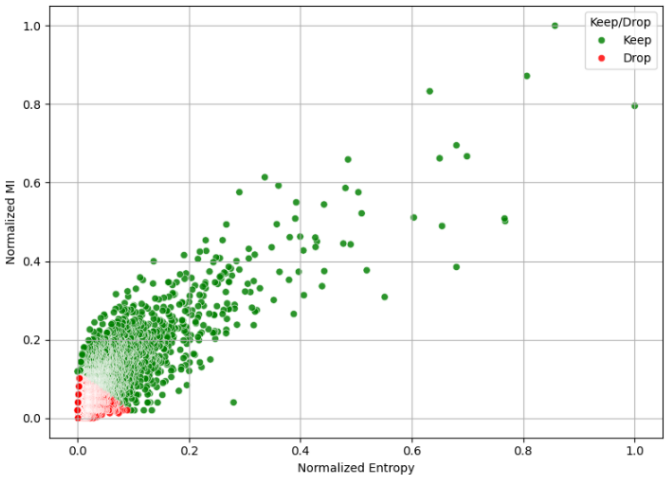


Fig. 8. Normalized MI vs Entropy (CORD-19).

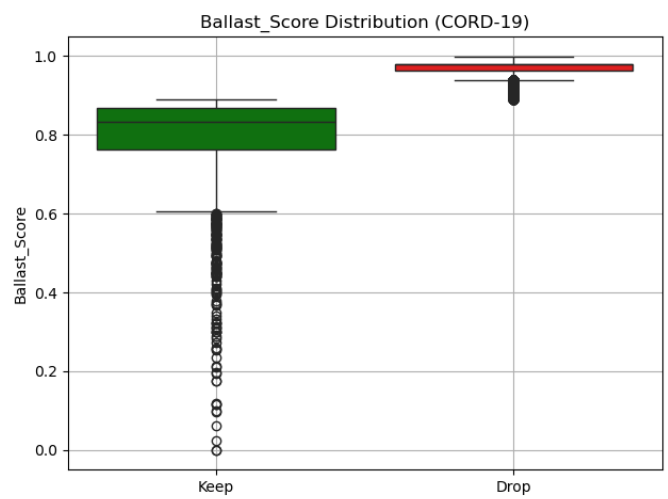


Fig. 9. Boxplot of ballast scores for retained versus dropped features in the CORD-19 dataset.

As illustrated in Fig. 10, the joint distribution of normalized mutual information and entropy reveals a clear separation between informative and ballast features in the PubLayNet dataset.

The scatterplot shows feature-level Keep/Drop annotations based on an entropy threshold of 0.5, highlighting how low-entropy and low-MI features cluster in the ballast region.

As shown in Fig. 11, dropped features have consistently higher Ballast_Scores, aligning with MI results that confirm their limited utility for layout analysis.

2) Semantic Redundancy via Topic Modelling and Embedding Similarity

Topic modelling (LDA) and BERT-based similarity analyses identified thematic ballast. In PubLayNet, captions and titles showed low Jensen-Shannon divergence, indicating semantic overlap. In CORD-19, LDA topics with coherence scores ($C_v < 0.3$) often reflected repetitive procedural text with low discriminative value. BERT cosine similarity confirmed that lexically varied but semantically similar content as e.g., standard methods sections, formed dense, low-utility clusters.

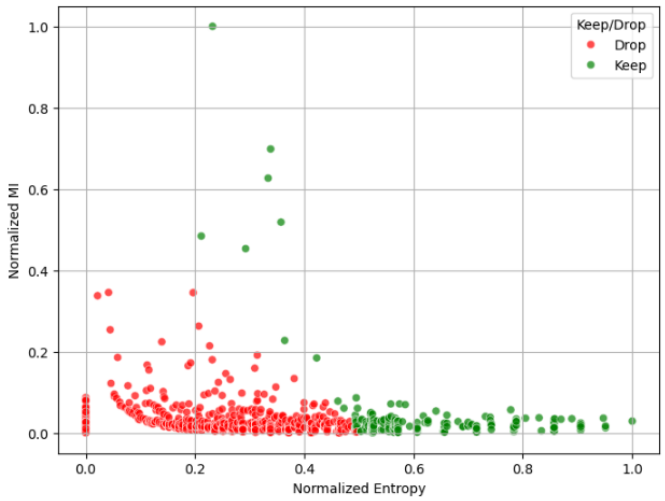


Fig. 10. Normalized mutual information versus entropy for PubLayNet features.

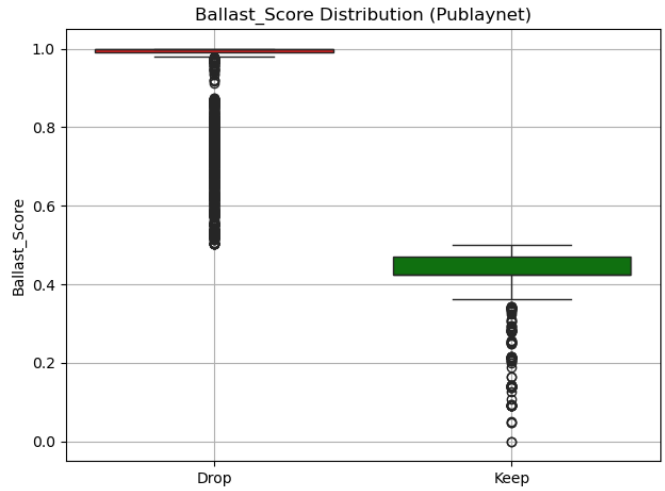


Fig. 11. Boxplot of Ballast Scores for retained versus dropped features in the PubLayNet dataset.

3) Comparative Evaluation of Ballast Detection Methods

Table III compares five ballast detection methods across CORD-19 and PubLayNet using peak metric values, corresponding thresholds, retained units, and observed behavioral trends.

Note: "**Peak Metric**" denotes the best-performing metric achieved by each method (e.g., coherence or retained information). "**Units Kept**" refers to the number of tokens or features retained at the peak threshold; lower values indicate more aggressive pruning.

This comparison highlights clear trade-offs, with some methods achieving strong performance under aggressive pruning, while others require higher feature retention to remain effective.

TextRank performs best without trimming in CORD-19 but degrades linearly as pruning increases, whereas LDA peaks at a moderate threshold of 0.30. BERT activates late in PubLayNet, peaking only at high thresholds (0.85+). Information Bottleneck shows non-monotonic trends in CORD-19 but improves steadily in PubLayNet. IoU-based trimming is aggressive early, with gains only at high thresholds. LDA is more stable in PubLayNet, though outperformed by TextRank in CORD-19.

Table IV summarizes the types of ballast identified by each method. Entropy and MI primarily detect structural and statistical redundancy and are effective across both datasets, while TF–IDF flags frequent but non-informative terms and LDA isolates incoherent topic blocks. BERT identifies semantically repetitive content, particularly in CORD-19, TextRank captures positionally generic phrases across datasets, and Information Bottleneck operates in a modality-agnostic manner by removing low-information elements while preserving utility. IoU-based methods primarily detect inter-document redundancy and are especially effective in PubLayNet.

These findings highlight method-specific strengths and their suitability across different data types.

C. Ballast Detection in Semi-structured Data

The `meta_Amazon_Fashion.jsonl.gz` dataset (826,108 records, 14 fields) exhibits a semi-structured, sparse format with nested structures, mixed types, and inconsistent field population. For example, `description` appears in only 7% of records and shows low entropy; `price` is present in just 6%. Features like `main_category` and `categories` are constant and thus identified as ballast. Numeric fields show low correlation and meaningful variance; `store` has high cardinality with no dominant class. Text fields such as `features` are dense but repetitive, necessitating dimensionality reduction. The heterogeneity of the data demands robust filtering using entropy, sparsity, and variance to isolate ballast from signal.

1) Ballast Typology in Semi-Structured Data

Mutual Information (MI): detecting statistical and model-irrelevant ballast

MI quantifies the dependency between features and a target variable (here, `average_rating`). MI revealed:

- `store_enc`: 0.1509 → high statistical relevance
- `features_tfidf`: 0.0404 → moderate relevance
- `price`: 0.0040 → negligible signal

The `price` field, while numerically valid, shows near-zero mutual information and is sparsely populated (∼6%), classifying it as statistical ballast. Fields like `main_category` and `categories`, previously identified as constant or empty, are also model-irrelevant ballast, offering no informational gain.

Entropy analysis: detecting statistical ballast in text

Entropy was applied to the `description` field to measure textual variability. As shown in Fig. 12, over 90% of records exhibit zero entropy, confirming that most descriptions are either missing or trivial. This lack of unpredictability indicates the presence of statistical ballast, even in a nominally rich textual field.

TF-IDF: exposing repetitive and redundant tokens

TF-IDF weighting of `description` reveals frequent, low-informational terms like "10", "cotton", and "machine." These generic size and material indicators dominate the vector space, suggesting semantic redundancy. While `features` retain

TABLE III
COMPARATIVE EVALUATION OF BALLAST DETECTION TECHNIQUES ACROSS DATASETS

Method	Dataset	Peak Metric (Value)	Threshold at Peak	Units Kept at Peak	Notes
LDA Coherence	CORD-19	0.3510 (Coherence)	0.30	26,349	Modest coherence increase followed by degradation beyond threshold 0.30
	PubLayNet	0.1648 (Mean Coherence)	0.25-0.85	14	Stable plateau after 0.25 with no additional gain
TextRank	CORD-19	0.9922 (Avg. Coherence)	0.00	28,608	Best performance without trimming; coherence degrades monotonically
	PubLayNet	1.5391 (Mean Rank Score)	0.80-0.85	6	Sharp improvement only at very high thresholds
Embedding BERT	CORD-19	0.2653 (Semantic Similarity)	0.45	15,792	Peaks at mid-threshold, followed by gradual decline
	PubLayNet	0.9036 (Mean Similarity)	0.85	2	Activates only under aggressive pruning
Information Bottleneck	CORD-19	1.0385 (Residual Info)	0.30-0.35	66	Fluctuating behavior; optimal at mid-range thresholds
	PubLayNet	0.7857 (Mean Info Retention)	0.85	6	Gradual improvement with a sharp jump at 0.85
IoU-Based	CORD-19	0.0781 (Mean IoU)	0.00	16,691	IoU drops sharply as threshold increases
	PubLayNet	0.6429 (Mean IoU)	0.80-0.85	14	Low retention until very high thresholds

TABLE IV
TYPOLOGY OF BALLAST INFORMATION IN UNSTRUCTURED DATA

Method	Ballast Type Detected	Description	Dataset Sensitivity
Entropy	Low-information tokens	Repeated headers, titles, and common layout blocks	Effective in both (stronger in PubLayNet)
Mutual Information (MI)	Statistically irrelevant text	Citations, method templates, and standard disclaimers	Effective in both (stronger in CORD-19)
TF-IDF	Lexically redundant terms	High-frequency but non-informative words	Effective in both
LDA	Topic-incoherent blocks	Clusters of incoherent or weakly defined topics (e.g., boilerplate methodological text)	Strong in CORD-19
BERT Embeddings	Semantically similar segments	Sections with low novelty or semantically repetitive content	Strong in CORD-19
TextRank	Positionally generic sentences	Frequently occurring mid-ranking or introductory statements with limited content value	Effective in both (especially textual data)
Information Bottleneck	Information-preserving compression	Tokens or features with minimal contribution to retained information	Strong in both (especially structured and semi-structured data)
IoU-Based	Inter-document redundancy	Tokens or features exhibiting high overlap across documents (e.g., shared templates)	Strong in PubLayNet and structured text

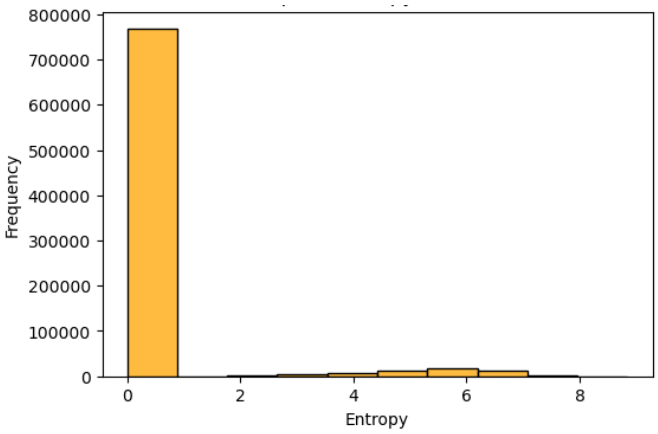


Fig. 12. Distribution of entropy values for product descriptions in the Amazon Fashion dataset.

useful signal, `description` exhibits repetitive ballast that is present, but with little discriminatory power.

LDA Topic Modelling: identifying low-coherence semantic ballast

LDA was applied to the `description` corpus. The resulting topics (e.g., "shoes, comfort, foot") are vague, high-level, and generic. Such broad thematic repetition marks the text as semantic ballast, where topical coherence is low and clustering offers minimal segmentation power.

BERT Similarity: detecting redundant semantic structures

Using BERT embeddings, pairwise title similarity was measured. A mean nearest-neighbour similarity of $\sim$0.66 indicates moderately high redundancy, suggesting many product titles encode overlapping semantics. This implicates `title` as redundant ballast, suitable for cluster-based deduplication.

Clustering (K-Means/DBSCAN): exposing redundant and noisy data

Clustering applied to BERT-encoded titles yielded $\sim$10 clusters with mixed densities (Fig. 13). Dense clusters reflect semantic duplication as e.g., nearly identical dresses with minor variations, while sparse clusters or outliers may signal noisy or irrelevant ballast.

Fig. 13 illustrates (top) the mutual information (MI) of selected features as `store_enc`, `features_tfidf`, and `price`, with the target variable `average_rating`, and (bottom) the distribution of 10 clusters from K-Means applied to title embeddings. `store_enc` has the highest MI (0.15), indicating strong predictive value, while `price` contributes minimally. The pie chart shows a balanced cluster distribution, suggesting diverse semantic patterns in the title embeddings, which is beneficial for downstream classification or segmentation.

Dimensionality Reduction (PCA / t-SNE): testing compressibility

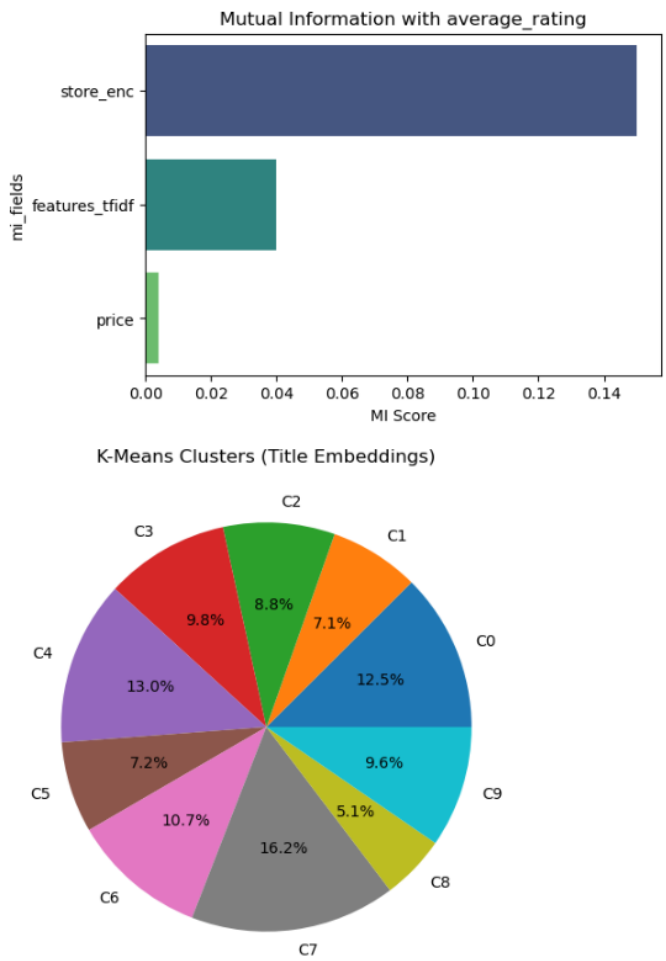


Fig. 13. Visual diagnostics of feature informativeness (top) and K-Means cluster distribution over title embeddings (bottom).

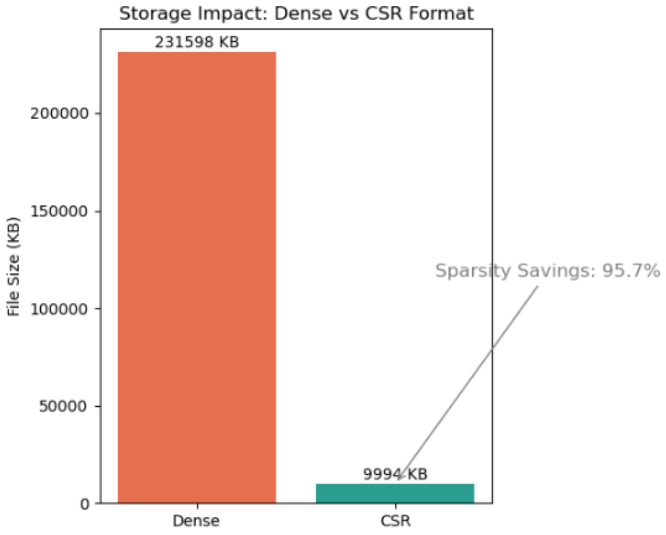


Fig. 14. Comparison of storage size for the TF-IDF vectorized "`description`" field using Dense and CSR formats.

TABLE V
SUMMARY OF THE METHOD–TYPOLOGY ALIGNMENT

Method	Ballast Typology Detected	Affected Fields
Mutual Information (MI)	Statistical, Model-Irrelevant	price, main_cat., categories
Entropy	Statistical	description
TF-IDF	Redundant, Semantic	description, features
LDA	Semantic	description
BERT Similarity	Redundant	title
Clustering	Redundant, Noisy	title, features
PCA / t-SNE	Weakly Redundant	title
Storage Impact (CSR)	Infrastructural/ Physical Ballast	description

PCA on title embeddings shows that 10 components explain ∼30% of the variance, suggesting a high-dimensional space with some redundancy, but no single dominant ballast dimension.

Storage Impact Analysis: physical manifestation of ballast

Storage-level ballast was assessed by comparing dense vs. sparse formats for the TF-IDF matrix of description. Storing it as a dense matrix requires 231 MB, whereas CSR (Compressed Sparse Row) reduces this to ∼10 MB, achieving 95.68% storage savings (see Fig. 14). This physical footprint quantifies infrastructural ballast, reinforcing the need to compress or discard sparse, low-utility text.

The CSR format reduces storage size by 95.68%, demonstrating its effectiveness in handling high-dimensional sparse data.

As summarized in Table V, ballast in semi-structured data spans statistical (MI, entropy), semantic (TF-IDF, LDA, BERT), and infrastructural (storage size) domains. Single-method approaches are insufficient, as comprehensive ballast detection requires combining statistical filtering, semantic analysis, and storage-level evaluation.

2) Ballast Detection & Reduction

Building on the initial typological mapping of ballast in the `meta_Amazon_Fashion.jsonl.gz dataset`, a systematic reduction strategy was applied, combining entropy, mutual information, cosine similarity, and model-based feedback to quantify ballast severity and eliminate redundant, low-utility features.

Fig. 15 presents the entropy-MI pruning curve, highlighting performance trade-offs during progressive feature elimination. As thresholds increase, the feature count drops from 499 to fewer than 10, while LightGBM maintains stable accuracy (∼0.49) and F1 score (∼0.36) up to approximately 85% reduction. This extended plateau suggests that the majority of text-derived features are either semantically redundant or statistically inert.

This observation is reinforced in Fig. 16, where a large concentration of terms appears in the bottom-left quadrant (low entropy, low mutual information), identifying them as clear ballast candidates. While TF-IDF-based redundancy remains limited (Fig. 17), BERT-based similarity and topic modelling expose substantial semantic duplication, particularly within the *title* and *description* fields.

The integrated multi-signal pruning curve shown in Fig. 18 combines entropy, mutual information, topic coherence, redundancy, and information bottleneck signals to guide optimal

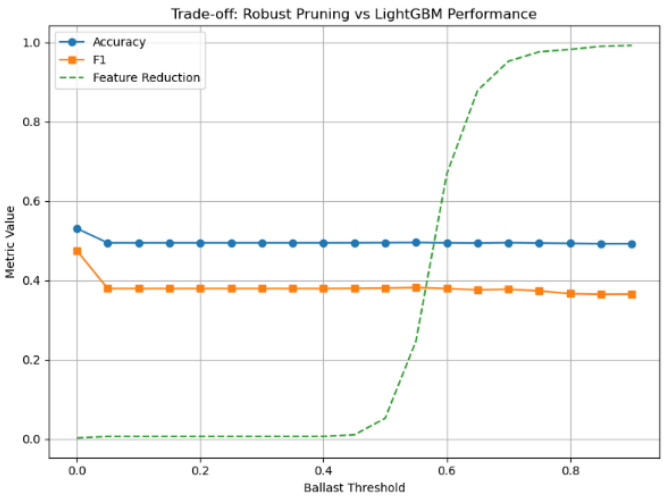


Fig. 15. Entropy-MI Pruning vs. Model Performance.

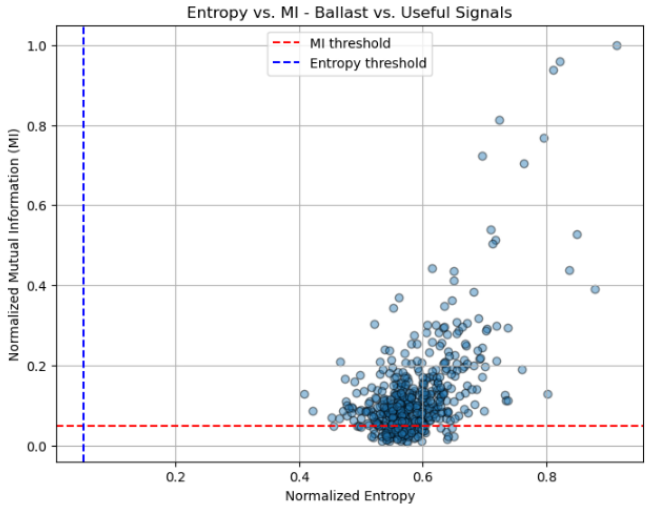


Fig. 16. Scatter Plot: Normalized Entropy vs. MI

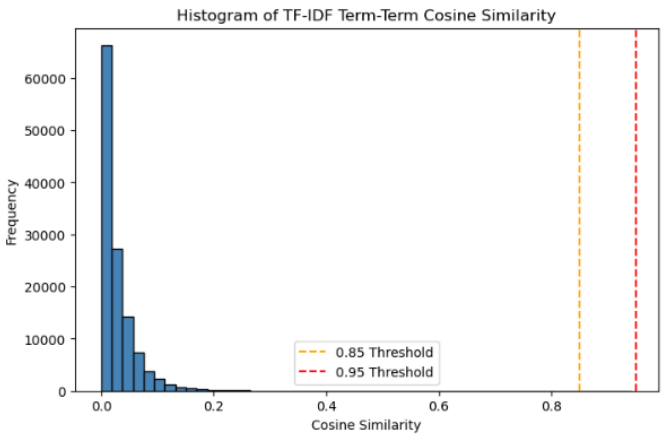


Fig. 17. TF-IDF term redundancy histogram.

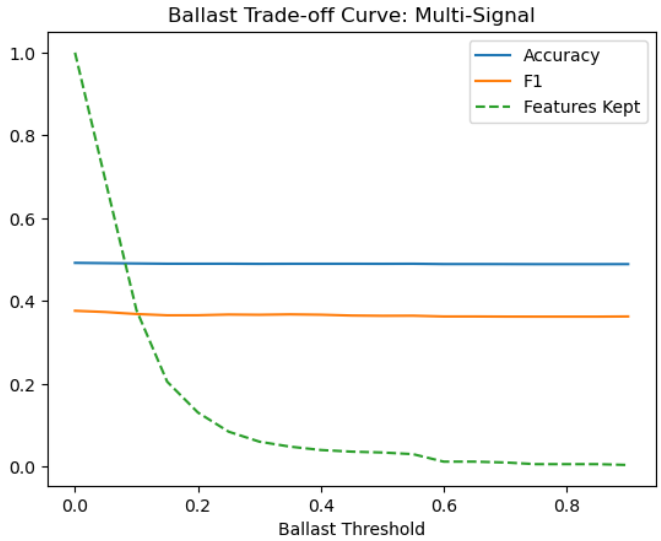


Fig. 18. Multi-Signal Ballast Trade-off Curve.

threshold selection. Beyond the 0.3-0.4 range, classification accuracy begins to decline slightly, indicating the transition from ballast removal to signal loss. Nevertheless, even at 99.4% feature reduction, model performance drops by only 0.004 from its peak value, while training time is reduced by nearly 50%.

These results confirm that ballast in semi-structured datasets is systemic in nature, encompassing statistical, semantic, and physical dimensions. The proposed masking framework effectively captures this heterogeneity, enabling robust and reproducible reductions with minimal information loss.

This curve demonstrates performance impact of pruning based on entropy and mutual information thresholds.

A quadrant-based diagnostic for identifying trivial vs. informative tokens.

Entropy shows variation, MI shows predictive relevance. Their intersection reveals true signal by filtering out both constant, low-variation terms and misleading, non-predictive noise. The entropy-MI plane makes the ballast visible.

Histogram shows co-occurrence redundancy is minimal, supporting semantic-level filtering.

This curve illustrates the trade-off between feature retention and model performance when applying increasing thresholds to multi-signal ballast pruning (TF-IDF, Mutual Information, Topic Coherence, Redundancy, and Information Bottleneck). As pruning threshold rises, the number of retained features drops sharply, while accuracy and F1 scores initially remain stable, then degrade.

D. Ballast Detection in Sparse Data

The sparse dataset, derived from 2022 Ireland Census, exhibits a high-dimensional structure (823 columns) with significant sparsity. After standard cleaning and integration, overall sparsity remained nearly unchanged (13.69% $\rightarrow$ 14.17%), indicating that conventional pre-processing fails to reduce ballast. Over 50 features exceeded 70% sparsity as some nearing 90%, suggesting low structural utility. These persisted despite deduplication and formatting corrections, highlighting structural ballast. A subset of sparse features was isolated for

TABLE VI
FEATURE REDUCTION STEPS AND REMAINING FEATURE COUNTS

Step	Description	Features Remaining
Raw Dataset	Initial feature set	47
Variance Threshold	Removed near-zero variance features	46
Correlation Filter	Dropped highly correlated features ($r > 0.95$)	42
Mutual Information	Retained features informative to the target variable	18
Entropy Filter	Confirmed relevance; no further reduction observed	18
LassoCV	Regularized linear feature selection	14
SHAP (Random Forest)	Retained features above mean SHAP importance	4

targeted reduction. No cardinality or variance pruning was applied at this stage, preserving redundancy and underscoring the need for advanced ballast-aware techniques.

1) Profiling Sparse Ballast

A focused analysis was conducted using both numerical (*T5_2_TP*) and categorical (*UR_Category*) targets. Each sparse feature was profiled for sparsity, uniqueness, variance, correlation (Pearson, Spearman), and mutual information (MI). Results showed consistently high sparsity (>70%) and low MI (<0.04), with negligible correlation to targets. Most features exhibited low variance and weak dependencies as clear indicators of ballast. Visualizations (heatmaps, scatter plots) confirmed that most features fell into the high-sparsity/low-signal quadrant. Only a few, such as *T5_2_GE8PH* and *T9_2_PI*, showed moderate informativeness. Some high correlations with *UR_Category* were tied to spatial or ID-like fields, raising leakage concerns. These findings affirm that sparse datasets often contain substantial ballast, requiring targeted dimensionality reduction.

2) Ballast Reduction Pipeline

A multi-stage pipeline was applied to a reduced sparse subset ($13{,}185 \times 47$). Initial variance and correlation filters removed static and redundant features. Mutual information and entropy filters preserved only features with meaningful signal diversity. LassoCV further pruned low-impact variables by shrinking coefficients to zero. Finally, SHAP (Random Forest-based) ranked the remaining features by model relevance.

As summarized in Table VI, this process reduced the original feature set by 91%, yielding a compact and informative subset of four features for downstream modelling.

Fig. 19 illustrates the average SHAP values of the 14 features retained after Lasso-based selection. Only four features exceed the mean SHAP importance threshold (highlighted in blue) and are therefore retained in the final reduced representation.

3) Ballast Detection & Pruning (Entropy + MI)

This phase employed a composite entropy-mutual information framework. Features were scored using a product-based Ballast Score and pruned across thresholds from 0% to 95%. CatBoost, selected for its robustness with sparse and categorical data, was used to evaluate classification and regression performance.

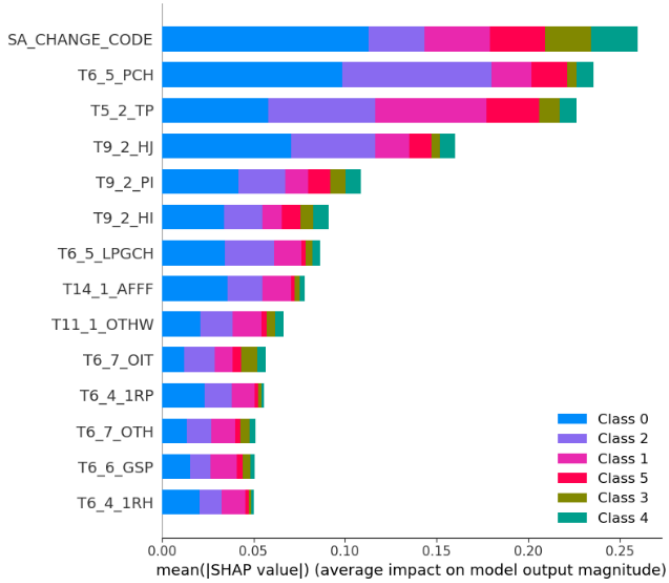


Fig. 19. SHAP feature importance bar plot.

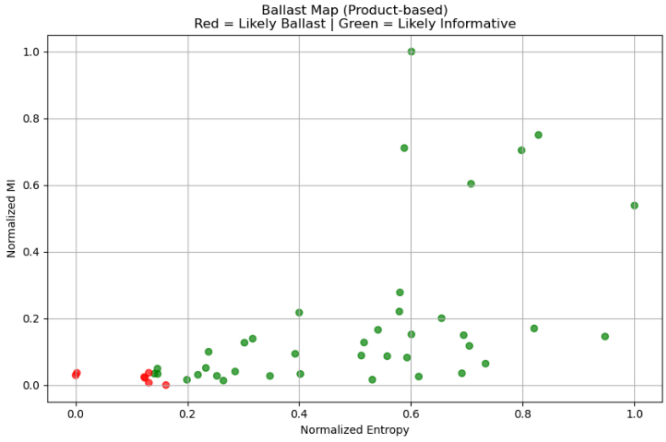


Fig. 20. Normalized entropy vs. normalized MI scatterplot, coloured by ballast score.

Fig. 20 visualizes the joint distribution of normalized entropy and mutual information, with features coloured by their ballast scores. A clear separation emerges between informative features and high-ballast candidates.

The resulting performance trade-offs are shown in Fig. 21, where predictive metrics improve significantly as pruning increases, peaking at approximately 85-86% feature removal. Beyond this point, over-pruning leads to a gradual degradation in generalization performance.

Performance stability across the critical threshold region is examined in Fig. 22. Stable behaviour is observed between thresholds 0.80 and 0.82, while threshold 0.84 marks a noticeable increase in AUC and recall. Based on these trends, a final pruning threshold of 0.86 was selected, retaining 39 high-value features and maximizing AUC (0.81) while minimizing model complexity.

The observed stability across aggressive pruning thresh-

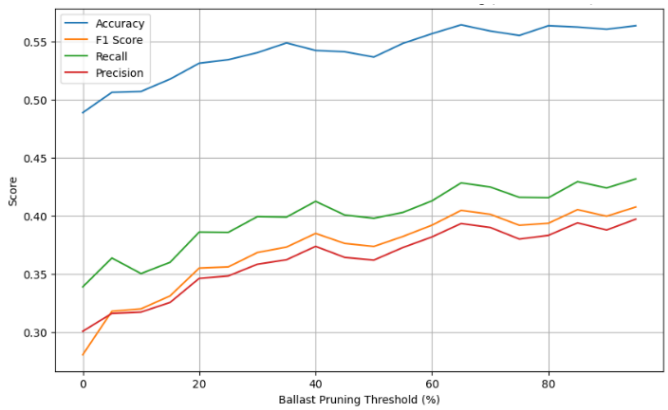


Fig. 21. Trade-off curve: performance vs ballast pruning threshold.

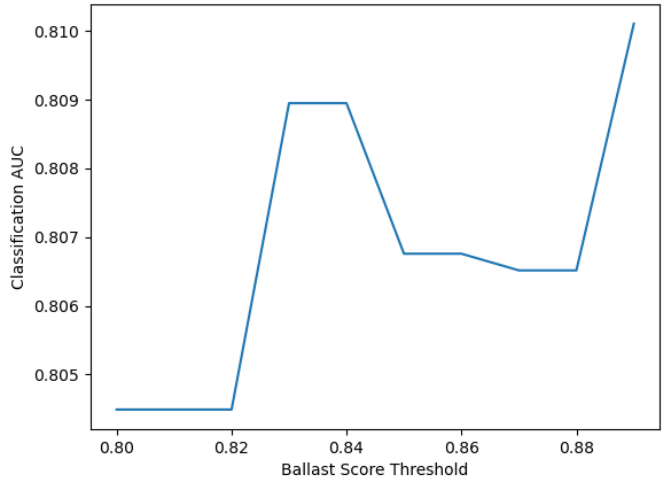


Fig. 22. Performance stability across ballast score thresholds between 0.80-0.89.

olds suggests that most features contribute marginal signal, reinforcing the need for multi-criteria ballast detection. This enables principled dimensionality reduction while preserving model robustness and generalization.

4) Ballast Detection & Pruning (PCA, VarianceThreshold, Correlation)

PCA exhibits a clear trade-off between dimensionality and predictive performance, as shown in Fig. 23. Retaining 85% of the variance (26 components) preserves a high AUC ($\sim$0.793), confirming that a substantial portion of ballast can be compressed without measurable performance loss.

VarianceThreshold filtering was effective up to thresholds of 0.10-0.14 (Fig. 24 - 25), while correlation-based pruning peaked at thresholds $\sim$0.44-0.46, retaining $\sim$35 features and achieving top AUC ($\sim$0.859, Fig. 26 - 27). Across methods, the optimal post-pruning feature count converged to 32-35, indicating a robust range for ballast-free, performant representations.

Fig. 23 shows how Random Forest performance metrics (AUC and Accuracy) vary with different levels of PCA variance retention. The red dashed line represents the baseline AUC using all 47 features (no dimensionality reduction). As retained variance increases, performance steadily improves,

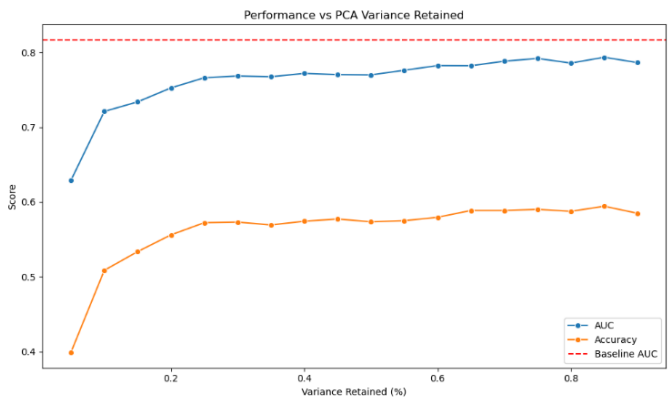


Fig. 23. Effect of PCA-based dimensionality reduction on Random Forest performance.

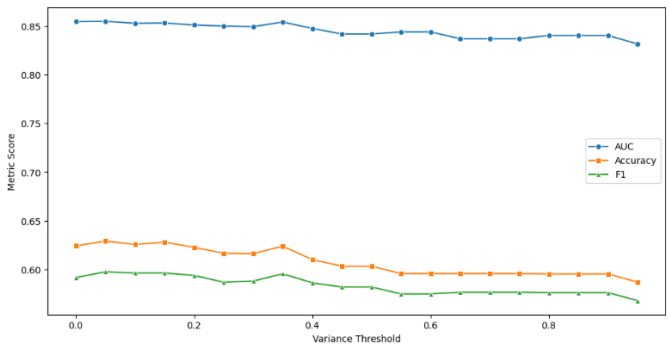


Fig. 24. Variance Threshold: Metrics vs. Threshold.

with a near-optimal AUC reached at 85% variance retained (26 components).

Fig. 24 displays how AUC, accuracy, and F1 score change as features with increasing variance thresholds are removed. Performance remains stable in the 0.05-0.15 range but declines beyond 0.20, suggesting valuable features begin to be excluded.

Number of features (Fig. 25) retained decreases gradually with rising variance threshold. Optimal trade-off is seen around threshold 0.10-0.14, retaining 33-44 features.

AUC, accuracy, and F1 (Fig. 26) improve with correlation thresholds up to 0.44. Best performance is achieved around 0.42-0.46, indicating effective removal of redundant features without losing signal.

At correlation thresholds ≥ 0.40, approximately 35 features (Fig. 27) remain, corresponding to peak model performance. Below 0.30, too many features are removed, harming predictive power.

5) Ballast Detection & Pruning (Lasso, SHAP, and Information Bottleneck)

L1-based pruning revealed a sweet spot at $\alpha \approx 0.001$, retaining $\sim$34-41 features with minimal AUC loss ($\sim$0.85). SHAP pruning with CatBoost showed stable performance up to 50% feature removal, peaking at AUC$\approx$0.805 at a threshold of 0.01 (Table VII). Finally, IB pruning identified ballast via normalized mutual information, yielding optimal classification (AUC$\approx$0.81) and regression performance when

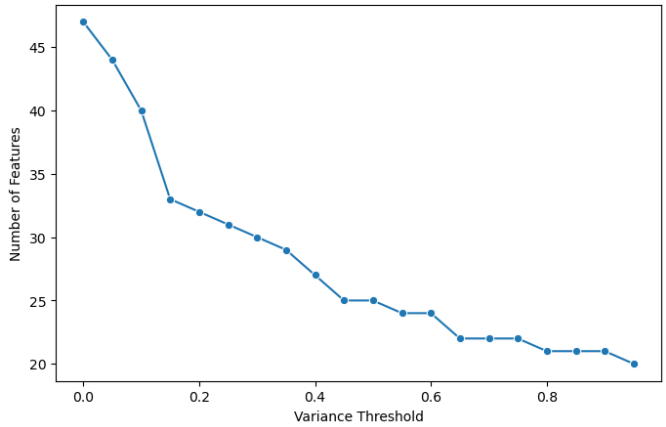


Fig. 25. Variance Threshold: Features Remaining vs. Threshold

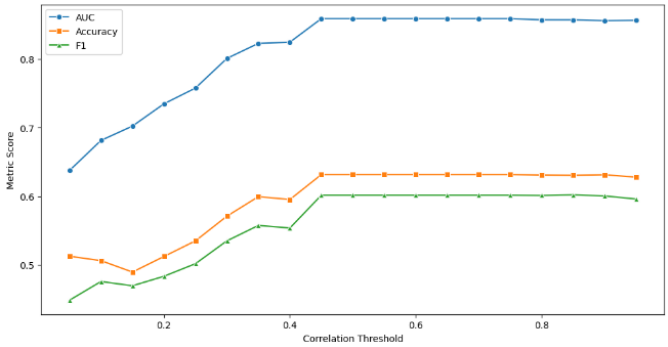


Fig. 26. Correlation Filter - Metrics vs. Threshold

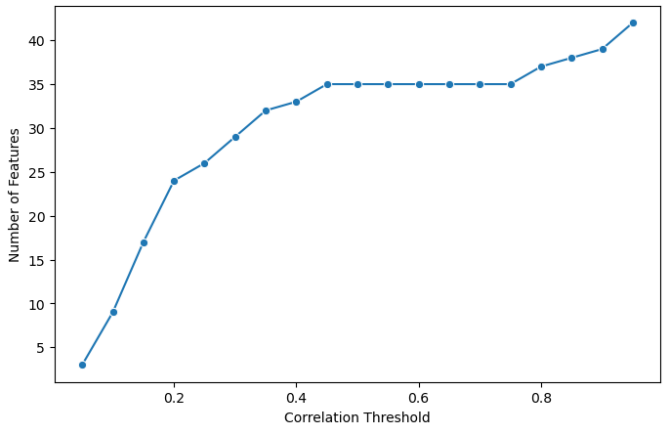


Fig. 27. Correlation Filter - Features Remaining vs. Threshold.

TABLE VII
COMPARISON OF SHAP-BASED FEATURE SELECTION: LIGHTGBM VS. CATBOOST

Aspect	LightGBM	CatBoost
AUC (All Features)	∼0.74	∼0.80
AUC (After Pruning)	∼0.74 (best at 21-28 features)	∼0.80-0.81 (best at ∼22 features)
F1 Score	∼0.77	∼0.78
Precision / Recall	∼0.81 / ∼0.81	∼0.82 / ∼0.82
SHAP Feature Stability	Moderate fluct. under pruning	Stable up to ∼50% feature removal
Feature Importance Spread	Sharp drop-off after top-ranked features	More inform., smoother top-feature distrib.

37-41 informative features were retained. All methods confirm that judicious ballast reduction can streamline feature space without compromising predictive power.

Table VII presents a comparative analysis of SHAP-based feature pruning in LightGBM and CatBoost models. The evaluation includes AUC, F1 score, Precision/Recall, and qualitative assessments of feature importance distribution and pruning stability. The results indicate that CatBoost exhibits superior robustness and interpretability under aggressive feature pruning.

The advantages of the entropy + MI method include balancing sparsity awareness (entropy) and target informativeness (MI), providing interpretable pruning via two-dimensional scatter plots and score thresholds, exhibiting a smooth and stable performance curve across pruning thresholds (0.80-0.89), and offering easily customizable thresholds.

6) Typology of Ballast Information in Sparse Structured

Classification of ballast types in sparse datasets along with their characteristic indicators and corresponding detection or pruning methods, is presented in Table VIII. Each ballast category aligns with a specific set of statistical, structural, or model-aware techniques suitable for effective reduction.

E. Mathematical Model of Ballast Information in Multimodal Datasets

1) Ballast definition and core hypotheses

Let $D = \{X, y\}$ be a multimodal dataset with n samples and m features or data elements, where $X \in \mathbb{R}^{n \times m}$ and $y \in \mathcal{Y}$ is a task-dependent target (classification, regression, or clustering). In this framework, *ballast* refers to features or tokens that are structurally present but contribute *minimal analytical value*. Based on the multimodal study, four primary ballast types are distinguished:

- *Statistical Ballast*: Elements with low variance or entropy, indicating uniformity or noise.
- *Redundancy Ballast*: Elements strongly correlated or semantically overlapping with others, offering duplicated information.
- *Semantic Ballast*: In unstructured or semi-structured data, elements with low thematic coherence or low embedding information density.
- *Model-Irrelevant Ballast*: Features with negligible influence on model predictions (e.g., low SHAP value, Lasso zero-coefficient).

Hypothesis: Ballast arises from combinations of low utility (informativeness) and high redundancy and varies by modality and task. A general mathematical framework must normalize and integrate these signals to support robust, cross-modal ballast detection.

2) Ballast Index (Feature-Level)

The *BallastScore* is defined for each feature or token X_j as a

TABLE VIII
TYPOLOGY OF BALLAST INFORMATION IN SPARSE DATASET

Ballast Type	Description	Indicators	Detection / Removal Methods
High-sparsity features	Features with extreme missing or zero values	Sparsity >70–90%; many nulls or default encodings	Sparsity Filter, Entropy Filter
Low-variance features	Features with nearly constant values across records	Variance near zero; no discriminative power	VarianceThreshold
Redundant features	Features highly correlated with others (linear or monotonic)	Pearson/Spearman $\rho > 0.95$	Correlation Filter
Low mutual information	Features weakly associated with target labels	MI $<$ 0.03-0.05	Mutual Information (MI), Entropy–MI Scoring
Structural artifacts	System-generated identifiers or placeholders	Uniform patterns, IDs, spatial zones without predictive value	SHAP, Domain Filtering, Data Leakage Checks
Over-encoded categories	Sparse one-hot or multi-hot encodings of categorical fields	High dimensionality with sparse representation	PCA, Embedding, SHAP, Lasso
Contextually irrelevant data	Features unrelated to the modelling task (e.g., metadata, survey structure)	No logical link to target; fails feature importance tests	LassoCV, SHAP, Information Bottleneck (IB)

weighted sum of its utility deficits and redundancy indicators:

$$\textit{BallastScore}_j = \sum_{k=1}^{K} w_k \cdot (1 - U_{jk}) + \sum_{r=1}^{R} \lambda_r \, R_{jr}, \quad (9)$$

where:

- $U_{jk} \in [0, 1]$: normalized utility signals (lower values imply lower informativeness),
- $R_{jr} \in [0, 1]$: normalized redundancy metrics (higher values imply stronger redundancy),
- w_k, λ_r: scalar weights reflecting the importance of each signal,
- K: number of utility metrics (e.g., mutual information, entropy, variance, topic coherence, SHAP),
- R: number of redundancy metrics (e.g., correlation, cosine similarity, embedding overlap).

Interpretation:

- A high *BallastScore* (closer to 1) indicates low utility and/or high redundancy and thus represents a strong ballast candidate.
- A low *BallastScore* (closer to 0) implies that the feature is both informative and non-redundant.

This formulation supports features from structured (tabular), semi-structured (e.g., JSON, XML), unstructured (text or image), and sparse datasets by normalizing all signals to the range $[0, 1]$.

3) Dataset-Level Ballast Index

The *dataset-level ballast index* $B(D)$ is defined as the fraction of features classified as ballast based on a threshold $\tau \in [0, 1]$:

$$B(D) = \frac{1}{m} \sum_{j=1}^{m} \mathbb{I}\left[\textit{BallastScore}_j > \tau\right], \quad (10)$$

where $\mathbb{I}[...]$ is the indicator function, and τ is an empirically calibrated threshold (e.g., based on inflection points in model performance, AUC curves, or SHAP drop-offs).

This index quantifies the *proportion of ballast within a dataset* and can be reported per modality or feature group to localize ballast hotspots.

4) Pruning Criterion

The ballast pruning operator Π_τ is defined as:

$$\Pi_\tau(X) = \{X_j \in X \mid \textit{BallastScore}_j \leq \tau\}, \quad (11)$$

that is, only features with a *BallastScore* below the threshold τ are retained. The value of τ is tuned based on task-specific model performance.

This formal model quantifies ballast through a modular, interpretable index integrating four core dimensions: *statistical insignificance, redundancy, semantic incoherence,* and model *irrelevance*. By enabling multi-signal detection and cross-modal generalization, it provides a principled basis for task-aware data reduction in large-scale, heterogeneous datasets.

V. CONCLUSIONS AND FUTURE WORK

This study presents a comprehensive, multimodal framework for ballast detection and reduction across structured, semi-structured, unstructured, and sparse datasets. By integrating statistical, semantic, and model-aware signals, a generalized ***Ballast Score*** is established, and its effectiveness is demonstrated across diverse data regimes. The key finding is that ballast corresponds to structurally present yet analytically low-utility data, which can be systematically quantified and pruned with minimal degradation to downstream model performance. In multiple scenarios, including SHAP-based pruning for structured data and entropy-mutual information (MI) filtering for sparse and unstructured data, performance gains and significant reductions in computational or storage overhead are achieved. Collectively, these findings confirm that ballast represents a quantifiable and removable form of data inefficiency rather than random noise.

A substantial share of ballast in datasets is believed to originate from unconscious human decisions during system design and data collection. When creating schemas or logging systems, humans rarely foresee all future analytical needs, leading to the inclusion of fields that are redundant, sparsely populated, or irrelevant. This cognitive gap as driven by overengineering, organizational habits, or short-term tasks, results in semantically and statistically low-value data. Over time, such features persist without re-evaluation, especially in complex systems with multiple stakeholders.

Practically, this research has direct implications for large-scale data processing, particularly in domains where high-dimensionality or semantic redundancy impairs model interpretability, computational efficiency, and storage scalability. Additionally, modality-specific ballast typology (e.g., statistical ballast in structured data, semantic redundancy in unstructured text, or infrastructural ballast in sparse formats) provides actionable insights in selecting appropriate pruning strategies.

However, the study is not without limitations. While the proposed ballast index generalizes across data modalities, its calibration (e.g., threshold τ) remains dataset- and task-dependent. Moreover, SHAP and Lasso underperform in semantically complex or lexically noisy settings, such as OCR text and scholarly corpora, where signal is not linearly separable. Furthermore, although entropy and MI perform robustly in unstructured contexts, their class-agnostic nature may omit low-frequency but high-value features in imbalanced tasks.

Future work should focus on two directions. First, extending ballast framework to non-textual modalities such as images, graphs, and audio, where redundancy and informativeness manifest differently, would validate its generality. Second, dynamic or adaptive thresholding strategies for the Ballast Score, perhaps informed by performance plateaus, SHAP-gradient inflection or by some other method, could yield more robust and context-aware pruning.

Brief summary of the study:

1. Different methods reveal different facets of ballast.
2. Ballast is task-dependent.
3. Entropy and MI are reliable starting points. The entropy-MI plane makes the ballast visible.
4. Semantic pruning adds interpretive depth, but not always practical.
5. Reducing ballast improves more than just performance, it improves the entire data science workflow.
6. Evaluation needs multi-faceted metrics, including metrics like feature count reduction, training time, and post-pruning interpretability to give a fuller view.

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Appendix A
Unstructured Dataset (CORD-19): Detailed Methodological Description

A. Dataset Description and Justification

The unstructured data analysis was conducted using the CORD-19 dataset, an open-access corpus of scholarly literature related to COVID-19 compiled by the Allen Institute for AI. The dataset contains over one million scientific articles, including approximately 370,000 documents with full-text availability in JSON format. Available metadata includes titles, abstracts, publication information, and structured full-body sections.

The CORD-19 dataset was selected for the following reasons:

Semantic complexity: As a full-text corpus of academic publications, the dataset exhibits diverse linguistic structures, domain-specific terminology, and substantial variation in document length, representing core challenges of unstructured textual data.

Volume and variability: With hundreds of thousands of heterogeneous documents ranging from abstracts to full-length articles exceeding 1,000 sentences, the dataset is representative of real-world large-scale natural language corpora.

Scientific relevance: CORD-19 has been widely adopted in biomedical text mining and information retrieval research, providing a well-established benchmark for methodological evaluation.

Data modality: Unlike structured tabular data, information is encoded in free-form prose, making the dataset particularly suitable for evaluating semantic ballast detection methods based on redundancy, topical irrelevance, and entropy degradation.

B. Data Processing and Analytical Strategy

The ballast detection pipeline for unstructured data combined shallow statistical heuristics with deep semantic modelling techniques to identify low-information, redundant, or semantically uninformative textual units (ballast sentences) that reduce dataset utility without contributing semantic value.

The pipeline comprised the following stages:

1) Data Extraction and Sampling: Full-text JSON files were parsed, and the following fields were extracted: *title*, *abstract*, and *body_text*. Articles containing fewer than 100 words in the full-text body were excluded. For computational tractability, a stratified sample of approximately 10,000 documents was selected for analysis.

This sampling strategy ensured coverage of diverse document structures while maintaining manageable computational requirements. Short documents were excluded to avoid unstable or misleading entropy and redundancy estimates.

2) Text Pre-processing: Standard natural language processing pre-processing steps were applied:

- Lowercasing and stop-word removal using `NLTK`,
- Tokenization and sentence segmentation via `word_tokenize` and `sent_tokenize`,
- Lemmatization using `spaCy`,
- Removal of numeric tokens and punctuation-only tokens.

These steps reduced lexical noise and normalized textual representations to enable consistent vectorization and semantic modelling across documents.

3) Statistical Ballast Detection (Shallow Methods): An initial round of ballast identification employed statistical indicators of information sparsity.

a) *Term Frequency–Inverse Document Frequency (TF-IDF)*

TF-IDF scores were computed across the corpus using `TfidfVectorizer`. Sentences with aggregate TF-IDF values within the lowest decile of the distribution were flagged as low-information candidates.

b) *Shannon Entropy*

Sentence-level Shannon entropy was calculated based on empirical token frequency distributions:

$$H(S) = -\sum_{i=1}^{n} p(w_i) \log_2 p(w_i) \tag{12}$$

where $p(w_i)$ denotes the empirical probability of token w_i within sentence S. Sentences with entropy values below 1.5 were considered lexically sparse and marked as ballast candidates.

TF-IDF captures content distinctiveness, while entropy quantifies lexical diversity; both metrics are widely used indicators of information richness in text corpora [12], [22].

C. Semantic Ballast Detection (Deep Methods)

To refine the ballast filtering process, two semantically grounded techniques were employed.

1) Sentence Embeddings and Cosine Similarity: Sentence embeddings were generated using a Sentence-BERT model (DistilBERT), producing 768-dimensional vector representations. Pairwise cosine similarity was computed both between consecutive sentences and across entire documents. Sentences exhibiting cosine similarity values greater than or equal to 0.95 were flagged as highly redundant.

This step targeted semantic repetition, where sentences contribute minimal novel information despite lexical variation.

2) Topic Coherence via Latent Dirichlet Allocation: Latent Dirichlet Allocation (LDA) models were trained using Gensim with the number of topics set to 20. Each sentence was assigned a topic distribution, and Jensen–Shannon divergence was computed between sentence-level topic vectors and the dominant topic distribution of the corresponding document.

Sentences with high divergence values, indicating low topical coherence, were treated as off-topic ballast [23], [24].

D. Aggregation and Filtering Strategy

Outputs from the four ballast indicators (low TF-IDF, low entropy, high semantic redundancy, and low topic coherence) were aggregated using a voting-based decision rule. A sentence was classified as ballast if at least two of the four criteria were satisfied.

Filtered documents were reconstructed using only retained sentences, producing ballast-reduced textual representations for downstream evaluation.

E. Evaluation of Ballast Removal

To assess the effectiveness of ballast removal, a LightGBM classifier was trained to categorize documents into biomedical subdomains (e.g., virology, epidemiology) using both original and ballast-reduced text representations.

Evaluation metrics included Accuracy, F1-score, Recall, and ROC-AUC. Across experiments, ballast-reduced representations yielded modest but consistent improvements in classification performance, attributable to reduced semantic noise and redundancy.

Additional descriptive metrics included:

- Sentence reduction ratio (approximately 41% of sentences identified as ballast on average),
- Shifts in entropy and topic coherence distributions following filtering.

F. Implementation Details

The following libraries and tools were used:

- `pandas`, **numpy**: data manipulation and numerical processing,
- `NLTK`, `spaCy`: text pre-processing,
- `scikit-learn`: TF-IDF computation and evaluation,
- `sentence-transformers`: semantic embeddings,
- `Gensim`: LDA topic modelling,
- `LightGBM`, `SHAP`: downstream classification and interpretability,
- `matplotlib`, `seaborn`: visualization.

G. Summary

This *Appendix A* demonstrates that the proposed ballast detection framework can be systematically instantiated for large-scale unstructured text corpora by integrating statistical sparsity measures with semantic redundancy and topical coherence analysis. The presented configuration serves as a reference implementation for unstructured data modalities and complements the high-level methodology described in the main body of the paper.

APPENDIX B
BALLAST TAXONOMY TABLE (IX)

Table IX summarizes the comparative performance of ballast detection methods across structured, unstructured, semi-structured, and sparse datasets.

Interpretation

- *Structured Dataset:* Lasso, SHAP, and Correlation Filters were the most effective. PCA showed peak performance but with reduced interpretability. SHAP best balanced feature importance and interpretability. MI and Entropy provided early detection of non-contributory features.
- *Unstructured Dataset:* Entropy and MI outperformed others by identifying statistically and structurally redundant text. BERT and LDA provided complementary semantic ballast detection, while TextRank and IoU handled layout-based or templated ballast effectively.
- *Semi-Structured Dataset:* The best outcomes emerged from hybrid combinations as TF-IDF and LDA for semantic ballast, MI and Entropy for structural ballast, and SHAP for model-awareness. PCA was used cautiously to compress over-encoded metadata.
- *Sparse Dataset:* A multi-stage pipeline proved necessary. Entropy and MI led the reduction phase, followed by Lasso and SHAP for final selection. PCA and t-SNE were informative but not optimal for production modelling due to interpretability loss.

APPENDIX C
CONDENSED BALLAST TYPOLOGY TABLE (X)

For clarity, a condensed ballast typology is summarized in Table X.

TABLE IX
COMPARATIVE EFFECTIVENESS OF BALLAST DETECTION METHODS ACROSS DATA MODALITIES

Method	Structured Dataset	Unstructured Dataset	Semi-Structured Dataset	Sparse Dataset
Mutual Information (MI)	Detected statistically irrelevant features with weak target signal (e.g., price, category); rejected approximately 80 features.	Effective in isolating boilerplate and citation-like structures in CORD-19; also flagged low-content layout regions in PubLayNet.	Highlighted weakly informative textual fields and categorical encodings; effective for numerical metadata.	Rejected fields with MI < 0.03; highly effective for dimensionality reduction while preserving predictive signal.
Entropy	Detected flat distributions; flagged features with uniform value spread (e.g., device information).	Exposed repeated tokens and layout blocks in PubLayNet; revealed document templates in both datasets.	Identified repeated tokens in descriptions and metadata; reinforced MI-based filtering.	Detected zero-variance and ultra-sparse columns; strong synergy with MI for structure-first filtering.
SHAP	Ranked features by model relevance (LightGBM); four features retained post-processing; best interpretability–performance trade-off.	Performed poorly on text-dominant data due to limited structural grounding; weak thematic alignment.	Weak correlation with semantic methods; struggled with high-dimensional sparse textual inputs.	Final reduction stage; retained top four predictors based on SHAP values; excellent interpretability.
Lasso	Efficient linear pruning; eliminated zero-contributing fields ($105 \rightarrow 14$ features).	Missed nonlinear ballast in free-text; underperformed compared to semantic detectors such as LDA and BERT.	Detected structurally redundant fields; insensitive to lexical overlap; effective for numerical variables.	Effective after MI and entropy filtering; removed redundant encodings and noise features.
PCA / t-SNE	Filtered latent ballast; achieved best AUC (0.998) but sacrificed interpretability ($432 \rightarrow 23$ features).	Limited explainability for text; captured both noise and signal; effective only for embedding-based representations.	Detected over-encoded sparse fields; useful for visualizing low-dimensional semantic clusters.	Detected latent sparsity but removed interpretable signals; unsuitable for end-to-end model explanation.
Correlation Filter	Mitigated multicollinearity ($432 \rightarrow 116$ features) with minimal performance loss.	Not applicable to raw high-dimensional text without embeddings.	Detected duplicated structured metadata fields; ineffective for free-text descriptions.	Removed redundant population statistics using Spearman $\rho > 0.95$; foundational pipeline step.
TF-IDF	Not applicable.	Removed overused generic terms; effective for early-stage ballast detection in CORD-19 and PubLayNet.	Pruned non-informative tokens in textual descriptions; filtered spam-like reviews.	Not applied.
LDA (Topic Coherence)	Not applicable.	Detected topic-incoherent segments; optimal threshold at 0.30 for CORD-19; stable performance on PubLayNet.	Identified noisy review blocks lacking topic affiliation; effective for semantic ballast classification.	Not applied.
BERT Similarity	Not applicable.	Detected semantically repetitive content in CORD-19; later activation in PubLayNet (threshold ≥ 0.85).	Identified redundant or copy-pasted user reviews; correlated with embedding-based clustering.	Not applied.
Clustering (K-Means, DBSCAN)	Not applicable.	Identified text segments forming noisy, topic-independent clusters in both datasets.	Separated semantically weak clusters in review text; supported embedding-based filtering.	Not applied.
Information Bottleneck	Reduced features to 80; moderate AUC (0.72); effective as a compression-aware auxiliary method.	Captured low-contribution terms; non-monotonic gains; effective for layout-based ballast in PubLayNet.	Assisted MI and entropy methods; aided calibration of pruning thresholds.	Flagged features with residual contribution below $\tau = 0.30$; supported by MI and entropy rankings.
IoU-Based Redundancy	Not applicable.	Highly effective for PubLayNet (IoU > 0.64 at $\tau = 0.85$); flagged redundant layout components.	Weak signal; primarily useful for repeated field positions.	Not applied.

TABLE X
CONDENSED BALLAST TYPOLOGY AND CORRESPONDING DETECTION METHODS

Ballast Type	Detection Methods	Description	Effective In
Statistical Ballast	Entropy, Mutual Information (MI), Variance Threshold	Low-variability or uniform features with minimal discriminative power.	Structured, Sparse, Semi-Structured
Redundancy Ballast	Correlation Filter, Lasso, SHAP, PCA, IoU	Features duplicating others either numerically or structurally, such as collinearity or templated layouts.	Structured, Sparse, Unstructured
Semantic Ballast	TF-IDF, LDA (Topic Modelling), BERT Similarity, Clustering (K-Means, DBSCAN)	Lexically or thematically non-informative components, including boilerplate text, common phrases, and repetitive segments.	Unstructured, Semi-Structured
Model-Irrelevant Ballast	SHAP, Lasso, Information Bottleneck	Features exhibiting negligible contribution to model outputs, such as near-zero coefficients or low SHAP values.	Structured, Sparse, Semi-Structured